

YOUSSEF MOHARM

AI / Software Engineer | Computer Science Student

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PROFESSIONAL SUMMARY

Motivated Computer Science student at Alamein International University (expected graduation 2026) with hands-on experience in machine learning, data analysis, and backend software development. Proficient in Python, Java, and SQL, with practical exposure to scikit-learn, TensorFlow, NLP, and SOLID design principles. Completed a 6-month AI/ML bootcamp at the Information Technology Institute (ITI) and contributes to AI model evaluation as a freelance consultant. Active organizer within the IEEE AI community, mentoring junior members and leading technical workshops. Seeking an AI/software engineering role to apply machine learning and engineering skills to real-world problems.

EXPERIENCE

Freelance AI Consultant

2024 – Present

Outlier AI · Remote

- Analyzed and refined 500+ training datasets for machine learning model improvement, focusing on data quality validation and edge case identification
- Evaluated model outputs across computer vision, NLP, and reasoning domains, providing feedback to improve robustness and performance

AI and Machine Learning Trainee

Jan 2025 – Oct 2025

Information Technology Institute (ITI) · Remote · Student Training Bootcamp

- Completed a 6-month intensive AI/ML bootcamp with 500+ hours of hands-on training in Python, machine learning, and backend development
- Built a multilingual NLP chatbot handling 5+ languages with 92% accuracy using Python and advanced text preprocessing techniques
- Designed secure Java/Python backend systems applying SOLID principles and design patterns; optimized MySQL queries, reducing latency by 40%

AI Member & Organizing Committee Member

2025 – Present

IEEE Community · Alexandria, Egypt

- Organized and executed 8+ technical AI workshops and seminars, reaching 150+ students and professionals in the community
- Contributed to curriculum development within the IEEE AI department and mentored 20+ junior members in machine learning and software engineering best practices

PROJECTS

Smart City Transportation Network Optimization [\[GitHub\]](#)

2025

Python · Graph Algorithms · Dynamic Programming · Greedy Algorithms · Academic Project

- Architected a transportation management system for Greater Cairo using graph algorithms, dynamic programming, and greedy approaches
- Engineered MST algorithms for cost-efficient road design and Dijkstra's/A* search for emergency routing and traffic optimization

Credit Card Fraud Detection ML Model [\[GitHub\]](#)

2025

Python · Scikit-Learn · Machine Learning · Academic Project

- Built a complete machine learning pipeline for fraud detection on a dataset of 300K+ transactions, applying SMOTE for class balancing
- Trained and evaluated multiple models (SVM, KNN, Naive Bayes, Random Forest), achieving high precision and recall

Medical Diagnosis Expert System [\[GitHub\]](#)

2025

Python · Expert Systems · MYCIN Model · Academic Project (Team)

- Built a knowledge-based expert system using MYCIN-style certainty factors with a forward-chaining inference engine
- Programmed conflict resolution strategies (LEX, MEA, Complexity) to prioritize rule selection during diagnosis

NLP Preprocessing Pipeline [\[GitHub\]](#)

2026

Python · NLP · Scikit-Learn · Academic Project

- Constructed a text preprocessing pipeline for tweet analysis, including tokenization, stemming, and lemmatization
- Extracted features using word frequency analysis and TF-IDF vectorization to prepare data for sentiment analysis

Pac-Man Clone [\[GitHub\]](#)

2024

Python · Pygame · Game Development · Academic Project

- Created a Pac-Man clone in Python using Pygame with classic maze mechanics, pellet collection, and a dynamic scoring system
- Programmed AI pathfinding for ghosts using A* and semi-random patrol patterns, plus full collision detection and win conditions

Medical Image Enhancer [\[GitHub\]](#)

2025

Python · Streamlit · Image Processing · Academic Project

- Built a Streamlit-based web application to enhance medical images using image processing techniques aimed at supporting healthcare diagnostics
- Implemented features to improve image clarity and reduce noise, helping clinicians make better-informed diagnostic decisions

EDUCATION

Bachelor of Science (B.Sc.) — Computer Science

Oct 2023 – May 2026

Alamein International University · Alexandria, Egypt

- GPA: 3.0 / 4.0 — expected graduation May 2026
- Relevant Coursework: Design and Analysis of Algorithms, Database Management, Artificial Intelligence, Machine Learning, Data Structures, OOP

TECHNICAL SKILLS

Languages: Python, Java, C++, R, SQL, HTML5, CSS3, JavaScript

Machine Learning: Scikit-Learn, TensorFlow, Keras, PyTorch, Neural Networks, Supervised Learning, Model Evaluation

Data Science: NLP, Data Preprocessing, Feature Engineering, SMOTE, Data Analysis, Data Structures

Backend Development: Flask, RESTful APIs, OOP, SOLID Principles, Design Patterns, Database Design

Databases: MySQL, SQL Optimization, Query Tuning

Tools & Practices: Jupyter Notebook, Google Colab, Anaconda, Linux, Git, GitHub

Soft Skills: Problem-Solving, Critical Thinking, Collaboration, Leadership, Presentation, Adaptability

LANGUAGES

Arabic: Native speaker

English: Full professional proficiency — reading, writing, and communication

CERTIFICATIONS

Building Multi-Agent Systems — Microsoft

2026

Introduction to Agile Development and Scrum

2026

Introduction to Artificial Intelligence — IBM

2025

Introduction to Databases — IBM

2025

Engineering Practices for Building Quality Software

2026

Create Your First Automation Script Using Selenium and Java

2024

Object-Oriented Programming (OOP)

2025

Machine Learning

2025

Digital Systems: From Logic Gates to Processors

2024

+ 24 additional certifications — IBM · Microsoft · Coursera

2024 – 2026